East West University

Digital Logic Design (CSE345)

Assignment

Submitted by –

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Section: 05

Submitted to

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Topic: Design the BCD to Seven Segment display Decoder

Truth Table:

I_3	I_2	I_1	I_0	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1
1	0	1	0	x	x	x	x	x	x	x
1	0	1	1	x	x	x	x	x	x	x
1	1	0	0	x	x	x	x	x	x	x
1	1	0	1	x	x	x	x	x	x	x
1	1	1	0	x	x	x	x	x	x	x
1	1	1	1	x	x	x	x	x	x	x

K-Map for a:

I_3I_2 \ I_1I_0	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

Here, $\mathbf{a} = I_3 + I_1 + \sim I_2 \sim I_0 + I_2 I_0$

K-Map for b:

I_3I_2 \ I_1I_0	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	x	x	x	x
10	1	1	x	x

Here, **b** = $\sim I_1 \sim I_0 + \sim I_2 + I_1 I_0$

K-Map for c:

I_3I_2 \ I_1I_0	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	x	x	x	x
10	1	1	x	x

Here, **c** = $I_2 + \sim I_1 + I_0$

K-Map for d:

I_3I_2 \ I_1I_0	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	x	x	x	x
10	1	1	x	x

Here, **d** = $I_3 + I_1 \sim I_0 + \sim I_2 \sim I_0 + \sim I_2 I_1 + I_2 \sim I_1 I_0$

K-Map for e:

I_3I_2 \ I_1I_0	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	x	x	x	x
10	1	0	x	x

Here, **e** = $I_1{\sim}I_0 + {\sim}I_2{\sim}I_0$

K-Map for f:

I_3I_2 \ I_1I_0	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

Here, **f** = $I_3 + I_2{\sim}I_1 + I_2{\sim}I_0 + {\sim}I_1{\sim}I_0$

K-Map for g:

I_3I_2 \ I_1I_0	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

Here, **f** = $I_3 + I_2 + {\sim}I_2I_1 + I_1{\sim}I_0$